

Risk Assessment & Contingency Plan

Prevention, triggers, and recovery actions for the StoryTale implementation.

Assessment Approach

Risks are evaluated using qualitative probability and impact levels. High-impact risks receive an explicit prevention action, a measurable warning trigger, and a contingency that protects the core local reading experience.

10

tracked risks

7

high-impact risks

3

AI/API risks

Weekly

review frequency

Risk Priority Matrix

Probability	Medium Impact	High Impact
High probability	R4, R7	R3, R6
Medium probability	-	R1, R2, R5, R8, R9
Low probability	-	R10

Risk Register and Contingencies

ID	Risk	Prob.	Impact	Prevention / Mitigation	Contingency	Trigger
R1	Cloud API quota, outage, or slow response	Medium	High	Cache successful results; avoid duplicate calls; add retry and timeout rules.	Keep the last valid result and allow the user to retry later.	API error or quota warning
R2	Gemini story analysis returns inconsistent data	Medium	High	Use strict structured output, validation, stable IDs, and chapter-to-book context.	Reject invalid output and use a safe minimal chapter package.	Schema or continuity check fails
R3	Generated characters or assets change appearance	High	High	Lock reference images, character descriptions, reusable asset IDs, and approved catalogs.	Regenerate only the failed variant or replace it with an approved placeholder.	Identity score or visual review fails
R4	Offline voice packs make the app too large	High	Medium	Limit the first release to five optional, quantized ONNX voice packs.	Ship one default voice and let users download or remove other packs.	Install size exceeds target
R5	Animated Story Mode performs poorly on mobile	Medium	High	Pre-generate assets, cache audio, limit visible sprites, and profile on Android.	Reduce motion, image resolution, and active layers while keeping subtitles and audio.	Dropped frames or high memory use

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R6	Different EPUB structures break import or chapters	High	High	Normalize spine and HTML content; test multiple EPUB layouts and encodings.	Show manual chapter review and keep the original EPUB available.	Missing or merged chapters
R7	DeepL character allowance is consumed too quickly	Medium	Medium	Translate per chapter, cache every result, and display usage before a request.	Allow English-only reading until the allowance resets or is upgraded.	Usage reaches 80 percent
R8	Local data or generated assets are lost	Medium	High	Use durable local database and file storage with atomic saves and migrations.	Restore from a local export or rebuild only the affected derived assets.	Missing record after restart
R9	Schedule slips because the scope expands	Medium	High	Protect the MVP, review weekly priorities, and defer music and advanced polish.	Freeze new features and complete the reader, translation, audio, and one chapter demo.	A milestone misses its week
R10	Copyright, privacy, or API key exposure	Low	High	Use user-owned or public-domain EPUBs, local files, and server-side secrets.	Remove exposed credentials, rotate keys, and disable affected generated content.	Secret scan or content complaint

Response Rules

- Stop or isolate any change that threatens book data, API keys, or the ability to read an imported EPUB.
- Prefer a smaller working mode over a visually richer mode that crashes or blocks the reader.
- Record the trigger, decision, owner, and outcome in the weekly progress notes.
- Escalate any high-impact risk that remains open for two consecutive weekly reviews.
- Reassess probability and impact after mitigation is implemented.

Ownership and Review

The student developer owns the register and reviews it every week. Technical risks are checked during implementation and targeted validation; schedule and scope risks are checked against the Gantt chart. Any activated contingency becomes the next priority until the core StoryTale reading flow is stable again.